

Product Requirements Document (PRD)

Learning Management System (LMS)

Version: 1.0 **Status:** Approved for Development

1. Executive Summary

This Learning Management System (LMS) is a hybrid platform designed to manage the end-to-end lifecycle of physical training programs reinforced by mandatory digital gatekeeping. Unlike traditional e-learning platforms, this system creates a hard dependency between digital competency (prerequisite testing) and physical participation (classroom attendance).

1.1 Core Value Proposition

The system automates quality control for physical training by ensuring no student can occupy a physical seat without first proving digital competence. It manages limited physical resources (seats) via automated waitlists and bridges the offline gap using QR-based attendance tracking.

1.2 Key Objectives

- **Unified Management:** Centralize the administration of courses, instructors, students, and physical venues.
- **Competency Enforcement:** Prevent enrollment in physical courses until specific digital prerequisites are met.
- **Capacity Control:** Automate venue capacity management to prevent overbooking.
- **Hybrid Tracking:** seamless validation of physical attendance using digital tools (QR Codes), with support for offline environments.
- **Verifiable Credentialing:** Issue secure, publicly verifiable certificates upon successful completion of both digital and physical requirements.

2. User Roles & Personas

2.1 Administrator

- **Scope:** Full system access.
- **Primary Goals:** Maintain system integrity, manage course catalogs, oversee instructor assignments, monitor global analytics, and handle exception cases (e.g., manual overrides).

2.2 Instructor

- **Scope:** Assigned courses and cohorts.
- **Primary Goals:** Deliver physical training, track student attendance (even without

internet), grade subjective assessments, and monitor cohort progress.

2.3 Student / Learner

- **Scope:** Personal learning path.
- **Primary Goals:** Browse course catalog, pass prerequisite tests, secure a seat in physical training, access study materials, and obtain certification.

3. Functional Requirements

3.1 User Management & Authentication

- **Registration:** The system must support user registration with mandatory fields (Name, Email, Phone, Organization).
- **Authentication:** Secure login using standard protocols (JWT/OAuth2).
- **Role-Based Access Control (RBAC):** Strict segregation of duties. Admins must have the ability to promote/demote users and lock accounts.
- **Profile Management:** Users must be able to view their history, active enrollments, and earned certificates.

3.2 Course & Content Management

- **Course Creation:** Admins must be able to define:
 - **Metadata:** Title, Description, Category, Duration (e.g., "3 Days").
 - **Venue Details:** Physical location address and specific room number.
 - **Capacity Constraints:** Maximum number of seats available.
 - **Waitlist Cap:** Maximum number of allowed waitlisted users.
 - **Instructors:** Assignment of one or multiple instructors to a course instance.
- **Content Hosting:** The system must support the upload and rendering of:
 - PDF Documents (viewable in-browser).
 - Video Content (hosted or embedded links).
 - Presentation Slides.
- **Modular Structure:** Courses must support breaking content into "Modules."
- **Module 0 (Remedial & Foundation):**
 - A globally accessible module designed for foundational knowledge.
 - Must be indexed by topic tags to allow for targeted remediation recommendations.

3.3 Assessment Engine (Testing)

- **Question Bank:** Admins must be able to create and manage a repository of questions.
 - **Types:** Multiple Choice Questions (MCQ).
 - **Tagging:** Each question must be tagged with a specific topic or competency area (e.g., "Safety", "Regulations").
 - **Difficulty:** Optional difficulty rating.
- **Prerequisite Testing (The Gatekeeper):**
 - Courses must have the option to link a mandatory "Prerequisite Test."
 - **Scoring:** Automated grading immediately upon submission.

- **Attempts:** The system must track the number of attempts and score history.
- **Encryption:** All test questions and answers must be encrypted at rest.
- **Smart Remediation:**
 - Upon failure, the system must analyze the user's incorrect answers based on question tags.
 - The system must provide specific recommendations pointing the user to relevant sections within "Module 0" rather than a generic failure message.

3.4 Enrollment & Capacity Management

- **Conditional Enrollment:** The "Enroll" action must be strictly conditional. It is disabled if the Prerequisite Test is not passed.
- **Seat Allocation:**
 - Upon a valid enrollment request, the system must check the current enrollment count against the defined Course Capacity.
 - If Current < Capacity: The user is enrolled, and a seat is deducted.
 - If Current == Capacity: The user is diverted to the Waitlist flow.
- **Waitlist Management:**
 - Users can join a waitlist if the course is full but the waitlist cap is not reached.
 - **Promotion:** If an enrolled user cancels, the system must either auto-enroll the next waitlisted user or notify them of the opening (configurable).
- **Entry Pass Generation:**
 - Upon confirmed enrollment (not waitlist), the system must generate a unique, time-bound Entry QR Code for the student.

3.5 Attendance & Physical Training Management

- **QR Code Scanning:**
 - Instructors must have a dedicated interface to scan student Entry QR Codes.
 - Scanning must validate the code against the specific course and date.
 - Successful scan updates the student's status to "Attended."
- **Offline Capability (Critical):**
 - The Instructor interface must function fully without an active internet connection.
 - Scanned data must be stored locally on the device.
 - The system must automatically synchronize attendance data with the central server once connectivity is restored.
- **Manual Override:** Instructors and Admins must have the ability to manually mark attendance for exception handling (e.g., broken phone).

3.6 Certification & Verification

- **Eligibility Logic:** The system must only generate a certificate when three conditions are met:
 1. Prerequisite Test Passed.
 2. Physical Attendance Verified (via QR or Override).
 3. Final End-of-Course Exam Passed.
- **Certificate Generation:**
 - Automated PDF generation including Student Name, Course Title, Date, and

Unique Certificate ID.

- **Public Verification:**
 - Certificates must include a QR code or URL.
 - This link must lead to a public, unauthenticated page on the LMS that validates the certificate's authenticity to third parties (employers/auditors).

3.7 Communication & Notifications

- The system must trigger automated notifications (Email/In-App) for:
 - Enrollment Confirmation (with calendar invite/Entry QR).
 - Waitlist Status Updates (Joined/Promoted).
 - Class Reminders (24-48 hours prior).
 - Certificate Availability.

4. Non-Functional Requirements

4.1 Performance & Scalability

- **Concurrency:** The system must support high-volume concurrent access (e.g., 500+ users starting a test simultaneously) without degradation.
- **Latency:** Test grading must be completed within 1 second of submission.
- **Capacity:** Architecture must allow scaling to support up to 10,000 active student profiles.

4.2 Security & Compliance

- **Data Protection:** All personal user data and test content must be encrypted in transit and at rest.
- **Session Management:** Secure session handling with automatic timeouts for inactivity.
- **Audit Logging:** Comprehensive logs of all critical actions (Grading, Enrollment, Attendance marking) must be accessible to Admins for dispute resolution.

4.3 Reliability & Availability

- **Offline First:** The Instructor module must be treated as an "Offline First" application to ensure training continuity in remote locations with poor connectivity.
- **Data Integrity:** Prevent race conditions during enrollment to ensure the number of enrolled students never exceeds physical capacity.

4.4 Usability & Accessibility

- **Device Support:**
 - Student Portal: Responsive Web (Mobile/Tablet/Desktop).
 - Instructor Portal: Mobile-Optimized for handheld scanning.
 - Admin Portal: Desktop-Optimized for complex management tasks.
- **UI Feedback:** Clear visual indicators for system status (e.g., "Syncing...", "Offline Mode", "You are #5 on Waitlist").

5. Reporting & Analytics

- **Course Metrics:** Enrollment numbers, Waitlist depth, Drop-off rates.
- **Student Performance:** Pass/Fail rates on specific tests and individual questions (to identify weak curriculum areas).
- **Instructor Performance:** Attendance rates per cohort and student feedback scores (if applicable).
- **System Health:** Sync logs and error rates from offline instructor devices.